**IV B.Tech I Semester**

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| 23A39701 | **AI FOR SOCIAL GOOD (Professional Elective-IV)** | **3** | **0** | **0** | **3** |

## Course Objectives:

1. To understand the role of Artificial Intelligence (AI) in addressing global social challenges aligned with the UN Sustainable Development Goals (SDGs).
2. To explore the application of AI in healthcare, education, environment, disaster management, and governance.
3. To analyze ethical concerns including fairness, accountability, transparency, and privacy in deploying AI systems.
4. To provide students with the ability to design and implement AI solutions tailored for real-world social good use cases.
5. To encourage interdisciplinary thinking and collaboration with stakeholders such as governments, NGOs, and research institutions.

## Course Outcomes:

1. Identify and analyze the potential of AI in solving societal problems aligned with SDGs through real-world case studies involving NGOs and global organizations.
2. Apply AI techniques in healthcare, including predictive analytics, NLP for clinical data, and AI-driven diagnostics, while considering privacy and ethical concerns.
3. Evaluate AI applications in environmental monitoring, disaster management, and sustainable development using techniques like satellite analysis and smart agriculture.
4. Demonstrate awareness of ethical and legal aspects of AI deployment by examining algorithmic fairness, transparency, and responsible AI guidelines.
5. Design and develop AI projects for social good using public datasets, stakeholder engagement, and responsible deployment strategies for low-resource settings.

## Unit I: Foundations of AI for Social Good

Introduction to AI for Social Good, The UN Sustainable Development Goals (SDGs) and the Role of AI, AI for Poverty Alleviation and Food Security, AI for Public Policy and Governance, Role of AI in Education and Digital Literacy, Overview of Social Impact Assessment using AI, Data-Driven Decision Making for NGOs, Case Studies: UNICEF, World Bank, and AI4D Initiatives

## Unit II: AI in Healthcare and Epidemiology

Predictive Analytics for Disease Prevention, AI Models for Medical Imaging and Diagnostics, AI for Drug Discovery and Personalized Medicine, Healthcare Chatbots and Virtual Assistants, Remote Monitoring and Wearable AI, Natural Language Processing for Clinical Texts, Case Study: AI in Pandemic Response (e.g., COVID-19), Privacy and Ethical Issues in Healthcare AI

## Unit III: AI for Environment and Disaster Management

AI for Climate Change Modeling and Monitoring, AI for Smart Agriculture and Crop Monitoring, Early Warning Systems for Natural Disasters, Satellite Image Analysis using Deep Learning, Predictive Maintenance for Environmental Sustainability, AI for Wildlife Conservation and Tracking, Energy Optimization using AI, Global Initiatives: AI for Earth, Google AI for Environment

## Unit IV: Ethics, Fairness, and Accountability in AI

Algorithmic Bias and Fairness in AI, Explainability and Transparency in AI Models, Ethical Frameworks and Guidelines (EU, IEEE, etc.), Responsible AI and Human-in-the-Loop Design, Privacy-Preserving Machine Learning Techniques, Trustworthy AI: Legal and Regulatory Aspects, Societal Impact Assessment and Reporting, Case Studies of Biased Algorithms and Remedies

## Unit V: Building AI Projects for Social Impact

Defining a Social Good Problem and Use Case Selection, Data Collection and Curation for Public Datasets, Design Thinking and Problem Mapping for AI, Model Training, Validation, and Evaluation Metrics, Scaling and Deploying AI Solutions in Low-resource Settings, Community Engagement and Stakeholder Involvement, Presentation of AI for Social Good Capstone Projects, Collaborations with Government and Non-Profits

## Text Books:

1. Ray, Shalini. *Artificial Intelligence for Social Impact*. Springer, 2022.
2. Varshney, Kush R. *Trustworthy Machine Learning*. Manning Publications, 2022.
3. Russell, Stuart J., and Peter Norvig. *Artificial Intelligence: A Modern Approach*, 4th Edition, Pearson, 2020.

## Reference Books:

1. Emmanuel Letouzé (Ed.). *Data for Development: An Introduction to the Data Revolution*.
2. Cathy O'Neil. *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy*. Crown, 2016.
3. Virginia Dignum. *Responsible Artificial Intelligence: How to Develop and Use AI in a Responsible Way*. Springer, 2019.
4. Ben Green. *The Smart Enough City: Putting Technology in Its Place to Reclaim Our Urban Future*. MIT Press, 2019.

## Online Courses:

1. AI for Social Good – edX (University of Washington)
2. **AI for Everyone – Coursera by Andrew Ng**
3. **Data Science and Machine Learning for Public Policy – World Bank Open Learning Campus**